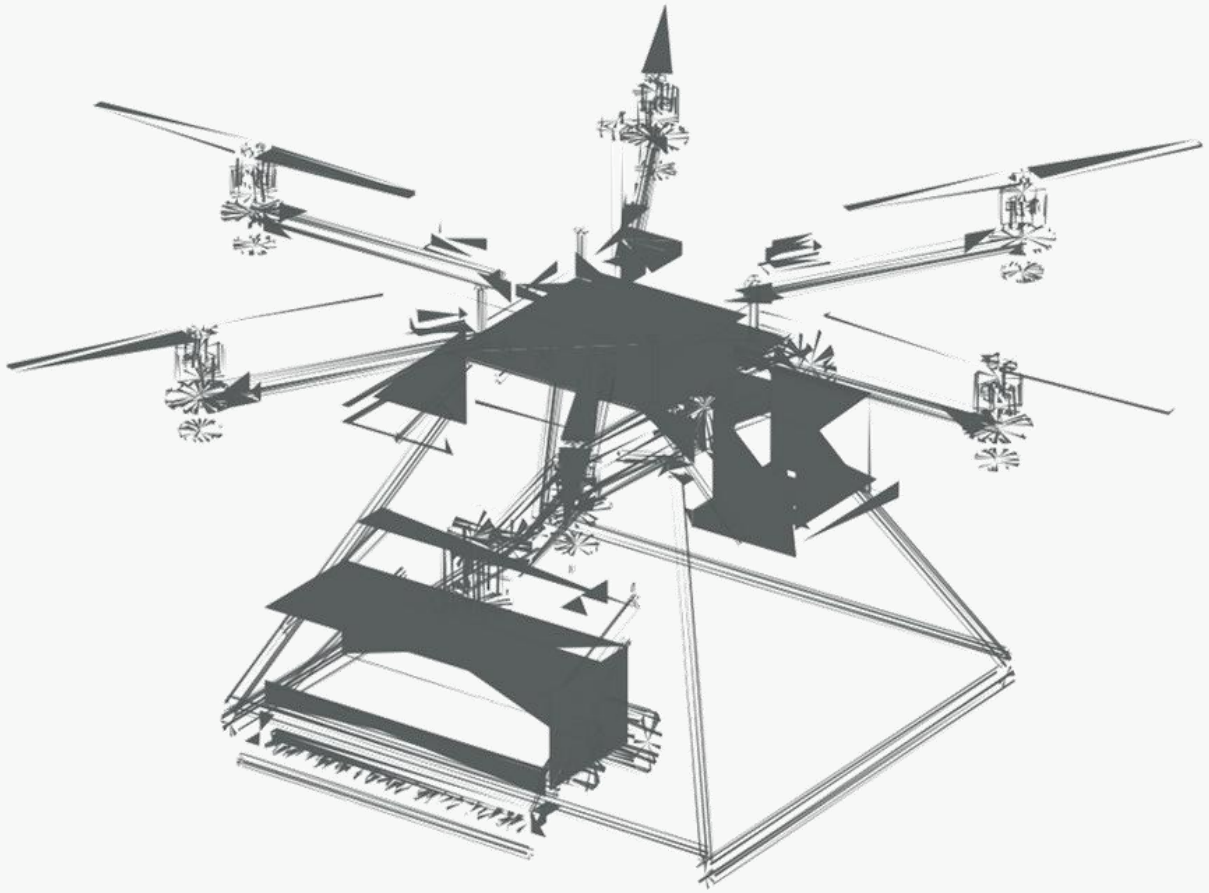
Current 3D assembly used for the study

Clean view of the latest complete drone concept: 6-arm frame + linear cutter + hose-coupler position.



Geometry locked for this study

- Front direction is between two arms, not aligned with one arm.
- Motor-to-motor diameter is 1460 mm for realistic 24 in prop clearance.
- Drone carries cutter, duct and hose coupler only; vacuum/storage remain on the ground.
- Hose coupler is near CG to reduce torque from the flexible tube.
- Cutter is a removable module mounted independently from landing gear.

This is the model used for the mass and propulsion assumptions below.

Executive answer

Six motors are enough for the first prototype if the final mass is controlled and the vacuum/storage stay offboard.

~29 kg	58.2 kgf	~2.0:1	~4.8 kgf
planning AUW incl. contingency	theoretical max static thrust	T/W ratio at 29 kg	hover thrust per motor

- The six-motor version is technically reasonable now because the drone does not carry the harvested-tea tank or the high-power vacuum motor.
- The biggest risk is weight growth: cutter module, mounts, hose drag, cable/hose forces, and vibration.
- The design should stay mechanically ready for X12 only if bench tests or measured mass show insufficient reserve.

Decision rule

≤29 kg: continue with 6 motors
29–33 kg: verify with thrust + tether tests
>33 kg or high hose force: move to X12

Recommendation: start with 6 motors, but keep X12 upgrade space in the CAD.

Six motors vs X12 coaxial

X12 gives more margin, but it is not free; it adds mass, current, complexity and coaxial interference loss.

Metric	6-motor hex	12-motor coaxial X12
Motor count	6 × MN701-S	12 × MN701-S
Theoretical max thrust	58.2 kgf	116.4 kgf before losses
Effective thrust assumption	~58 kgf	~99 kgf after ~15% coaxial loss
Added mass vs 6-motor	baseline	~+3.0 to +4.0 kg
Wiring / current	simpler	much higher peak current and heat
Best use	first prototype	upgrade if mass or hose force grows

Technical interpretation

X12 is not the first answer; it is the upgrade path. Use it if mass grows above the 6-motor comfort zone or if the hose/tether force needs more control authority.

Why not X12 immediately?

Because the vacuum/storage are offboard. Starting with X12 may add weight and electrical complexity before proving that we need it.

Recommended configuration for next CAD/Gazebo milestone

Proceed with a 6-motor frame, while keeping the CAD mechanically ready for X12 if tests require it.

1. CAD freeze

Finalize frame hardpoints, motor mounts, battery trays, landing gear, cutter interface and hose coupler.

2. Gazebo model

Use simplified collision geometry, measured masses and a tether disturbance model for hose force.

3. Bench tests

Test cutter vibration, airflow, blockage, duct leakage and hose pulling forces before flight.

4. Flight decision

If AUW/hose force exceed margin, activate X12 upgrade path.

Configuration freeze for now

- 6× MN701-S KV135 class motors.
- 24 in P24×7.2 class propellers.
- 6× 12S 80A-class ESCs.
- 2× 12S 22000mAh 25C packs.

Mechanical constraints to keep

- Motor mounts leave space for future lower coaxial adapter.
- PDB/wiring paths leave reserve for X12.
- Cutter and hose coupler remain removable modules.
- CG stays adjustable with battery/rear module positions.

Testing and validation plan

The propulsion decision should be confirmed by data, not appearance.

Test	Purpose	Pass condition
CAD mass update	Replace estimates with measured masses	AUW ≤29 kg target
Thrust stand	Verify motor/prop/ESC thrust and current	Enough thrust at safe current/temp
Cutter bench test	Measure vibration and power draw	No dangerous vibration / overheating
Hose drag test	Measure force from tube at different lengths	Force can be compensated by FC
Gazebo simulation	Check stability with mass + hose disturbance	Stable hover and path following
Tethered low-altitude flight	Validate real control margins	Hover throttle and logs acceptable

Important: the decision is not only 6 vs 12 motors.

The real decision is whether the measured AUW, hose disturbance, cutter vibration and hover logs are inside the comfort zone of the 6-motor design.